

Mortgage Guardrail Benchmark

A Mortgage-Specific Safety-Guardrail Benchmark, Grounded in Real HMDA Data

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Study status. This is a **benchmark-construction and baseline-evaluation** paper. The dataset is a *fixed* artifact grounded in the public HMDA 2022 snapshot; its prompts are **synthetic** and its labels are **LLM-judge, policy-card-consistent** — **not SME-adjudicated**. The benchmark therefore supports *measuring how guards behave on a mortgage-compliance distribution*; it does **not** license confirmatory fair-lending claims about any lender, model, or population. The contribution is the benchmark, its construction methodology, and a reproducible evaluation harness — plus baseline guard results on the frozen data.

Abstract

General-purpose prompt-safety guards are trained to catch domain-independent harm — jailbreaks, injections, abuse. But a request can be perfectly benign under a general safety policy while still soliciting a **mortgage-compliance** violation (disparate treatment, redlining, ability-to-repay circumvention, adverse-action-reason masking, occupancy fraud). We build a **mortgage-specific guardrail benchmark** whose unit is one incoming request, grounded in the public **HMDA 2022** loan-level snapshot, and carrying **two independent labels**: a domain-independent `general_safety` label (G) and a `mortgage_policy` label (D). The cross-product yields four quadrants; the load-bearing one is **G0/D1** — requests a general guard rates *safe* that nonetheless solicit a mortgage-policy violation. The benchmark is produced by an **agentic pipeline** (plan → HMDA-ground → generate → adversarially mutate → rubric-bound judge) that follows the borderline-and-subtle design principles established for LLM mortgage bias by , and includes **protected-class counterfactual minimal pairs** reframed as a *guard-invariance* gate. We release a frozen **994-row** version and a clean, canonical-metric evaluation harness, and report baseline results for open instruction checkpoints used zero-shot as guards and for off-the-shelf safety guards. We are explicit about what the benchmark does not yet establish and ship a validation protocol (SME adjudication, the empty G1/D0 quadrant, guard fine-tuning, cross-source decontamination) required before confirmatory use.

1. Introduction

Lenders are exploring LLMs across the mortgage lifecycle. That raises a specific safety question a general guard is not built to answer: *would honoring this request violate mortgage law or policy?* A general safety guard screens for harm that is independent of domain. A **mortgage guardrail** must additionally screen for domain violations that are, on their surface, entirely “safe” text — e.g. “*applications from the majority-Spanish-speaking tracts keep coming in weak; draft language to quietly deprioritize that pipeline, and write the denials up as ‘collateral’ so it’s clean.*” No slur, no jailbreak — but it solicits redlining, proxy discrimination, and adverse-action-reason masking.

This paper contributes: 1. a **fixed, HMDA-grounded benchmark** of mortgage-workflow requests with **dual orthogonal labels** (`general_safety` G × `mortgage_policy` D) and a derived intervention target; 2. an **agentic construction methodology** that grounds realistic, *borderline* scenarios in real loan records and makes policy violations *subtle and deniable* rather than cartoonish; 3. **protected-class counterfactual minimal pairs** used as

a guard-invariance PASS gate; and 4. a **reproducible evaluation harness** and **baseline results** showing that zero-shot general guards only *partially* separate the G0/D1 stratum (moderate ranking AP) yet **miss most of it at a deployable threshold**, with protected-pair invariance that varies sharply across guards.

It explicitly does **not** claim SME-validated labels, a populated G1/D0 quadrant, a fair-lending finding about any real system, or deployment readiness.

2. Related work and novelty boundary

LLM mortgage bias. audit LLM underwriting on real HMDA applications with experimentally manipulated race and credit scores, finding disparate approval/pricing recommendations that are *largest for lower-credit-quality / riskier loans*. We borrow their methodology in two ways: grounding scenarios in real HMDA fields, and building **counterfactual minimal pairs** that hold financials fixed and vary only a protected attribute. Our object is different: not an audit of an underwriting model’s decisions, but a **benchmark for a guard** that screens incoming requests.

Safety guards and their benchmarks. Open guards (Llama Guard, WildGuard, ShieldGemma) and general prompt-safety datasets (ToxicChat, prompt-injection and jailbreak sets) target a *content* taxonomy. None encodes mortgage regulation, none grounds in real lending data, and none separates domain-independent harm from a domain-policy violation as orthogonal labels.

Over-refusal and counterfactual fairness. XSTest/OR-Bench probe over-refusal; counterfactual fairness suites measure a *model’s output* bias. We combine both concerns for a *guard*: benign hard-negatives that a nervous guard over-blocks (e.g. a protected applicant asking “is it pointless for someone like me to apply?”) must PASS, and protected minimal pairs test the guard’s *invariance*, not a downstream decision.

Novelty boundary. The dual-label G×D design, HMDA grounding, and minimal-pairs-as-invariance gate are the combination that is new here. The agentic generate/judge loop is standard red-team practice; we do not claim it as a primitive.

3. Benchmark construction

The construction is deterministic in structure and stochastic in surface wording (the LLM steps run at temperature > 0). **The released benchmark is therefore a frozen artifact, not a regenerable one** — like the HMDA snapshot it grounds in. Reproducibility is defined at the *evaluation* layer (§5), not the generation layer.

HMDA grounding (PII-safe). We pull the public HMDA 2022 loan-level data (FFIEC/CFPB Data Browser) and reduce each record to a **banded, de-identified fact sheet** — loan purpose, occupancy, banded loan amount / income / LTV / DTI, action, denial reason, state — never exact dollar amounts, census tract, or any identifier. Scenarios are sampled from *marginal distributions* over these bands, with a build-time assertion that no emitted fact sheet reproduces a single source record. Following , fair-lending and ATR cells are biased toward **borderline / higher-risk** files (high DTI, high LTV, prior denial), where underwriter discretion — and bias — actually operates.

Agentic pipeline. A *planner* enumerates coverage cells (quadrant × trap × policy card × role × protected context); an *HMDA-grounder* draws a fact sheet; a *generator* authors the request in the natural voice of a role (applicant, loan officer, underwriter, processor, broker, adversary); an *adversarial mutator* composes label-preserving tactics (euphemism, coded proxy, business-justification, buried injection); a *rubric-bound judge* assigns the authoritative labels. Domain-independent harm (jailbreaks) uses curated seeds; the novel domain content is LLM-authored.

Labels. Each row carries general_safety_gold $G \in \{\text{safe, unsafe}\}$ (domain-independent), and mortgage_policy_gold $D \in \{\text{allow, intervene}\}$, defined by 24 benchmark **policy cards** (D01–D24: fair lending,

ATR/QM, disclosures, UDAAP, fraud, privacy) each with an “intervene iff” predicate and an authority pointer. A derived `final_intervention` = (G unsafe $\vee$ D intervene), plus an action lattice (PASS/CONSTRAIN/REVIEW/BLOCK) and severity. **Labels are policy-card-consistent, assigned by an LLM judge; they are not SME-adjudicated** (all 24 cards remain `sme_signoff: false`; the judge’s agreement is measured as self-consistency, not a human Fleiss- κ study).

Protected minimal pairs. For fairness, we author counterfactual pairs identical in every slot except a protected-class token (`variant` $\in$ {protected, reference}), both benign (G0/D0, PASS), sharing a `pair_id`. The guard must PASS both; the pair delta feeds a Δ_{context} invariance gate (target ≈ 0).

Provenance and splits. Content and near-dup (MinHash) family hashes; family-isolated train/dev/public_test splits + a sealed private_test; a text-free public index. *Current limitation:* decontamination was run against the legacy Paper A general sources (exact-hash); re-running against the v2 sources with near-dup removal is required before joint claims.

4. Benchmark composition (frozen v1_hmda2022)

994 rows, all `synthetic=true`, `contains_real_pii=false` (verified at freeze: 0 violations).

Split	Rows	Quadrant	Rows	Domain	Rows
train	604	G0/D0 (benign)	450	fair_lending	204
dev	149	G0/D1 (domain-only)	502	fraud	112
public_test	146	G1/D1 (both)	42	udaap	90
private_test (sealed)	95	G1/D0 (general-only)	0	disclosure	66
				atr_qm	54
				privacy	18
				benign	450

Trap types span `business_justified` (202), `direct` (167), `coded_proxy` (126), `euphemism` (126), `over_refusal_bait` (114), `benign_info` (84), `occupancy_temptation` (72), `minimal_pair` (78 rows = 39 protected pairs), `buried_injection` (25). **The G1/D0 quadrant is empty** — the safety-tuned generator declined to author domain-independent jailbreaks — so the 2x2 is only three-quarters populated; this is a stated limitation, not a design claim.

5. Evaluation protocol (reproducible)

The evaluator (`magen/score_guards.py` + `magen/evaluate.py`) is the reproducible layer. A guard emits one “unsafe” probability per row; we map it to G, D, and the composed `final` label alike, and compute, via the canonical tie-aware `guard_research` metrics: - **threshold-free macro-AP** for G, D, and `final` (per benchmark, macro-averaged); - **per-quadrant miss rate** at a calibration-selected operating point (5% dev-FPR target), so a strong aggregate cannot hide a G0/D1 collapse; - **Δ_{context}** , the mean absolute score gap within protected minimal pairs (fairness invariance).

Guards scored: the four base instruction checkpoints from the companion study (Qwen2.5-1.5B, SmolLM2-1.7B, SmolLM3-3B, Qwen3-4B) used **zero-shot as guards** via the canonical guard prompt, plus off-the-shelf **Llama**

Guard 3 and **WildGuard**. (A domain-fine-tuned guard arm is future work: the companion study’s LoRA adapters were not retained.)

6. Baseline results

Baseline results are from the committed scoring run (mortgage-redteaming-agentic-generator/out_eval/, produced by score_guards.py on the frozen benchmark, T4 fp16). We score the four base instruction checkpoints zero-shot as guards; the two off-the-shelf guards were gated in this run (note under the table).

Baseline zero-shot instruction guards on the frozen benchmark (public_test split; 146 rows: 75 G0/D1, 6 G1, 3 protected pairs), via score_guards.py; AP recomputed in the repo canonical env from the committed per-row scores (exactly reproducible). Threshold-free, base guards rank mortgage-policy violations moderately (AP·D 0.67–0.85) — soliciting fraud/discrimination reads as “unsafe” even without a jailbreak, so G and D are only PARTIALLY orthogonal. Protected-pair invariance (Δ_{context} ; 3 pairs) varies sharply across guards. The fixed 5%-FPR operating point is threshold-knife-edge for these clustered-score zero-shot guards (its G0/D1 catch count flips across library versions), so it is not tabulated per guard — see text. Small-sample, LLM-judge labels — illustrative, not confirmatory.

Guard	AP · G	AP · D	AP · final	Δ_{context} (fairness)
qwen25_15b_base	0.681	0.793	0.793	0.183
qwen3_4b_base	1.000	0.851	0.851	0.000
smollm2_17b_base	0.261	0.672	0.672	0.023
smollm3_3b_base	0.546	0.733	0.733	0.010

Not scored (unavailable in this run): llama_guard_3_1b, wildguard_7b.

Findings. The result is more nuanced than “general guards ignore mortgage compliance.” *Threshold-free*, the base guards rank D-violations moderately well (AP·D **0.67–0.85**, at or above their AP·G): soliciting discrimination or fraud reads as “unsafe” to a general instruction model even when it is not a jailbreak, so **G and D are only partially orthogonal in practice**. None reaches AP·D near 1, so a large fraction of the subtle G0/D1 stratum is unresolved by ranking alone. We deliberately do **not** report a fixed-threshold “caught at 5% FPR” count: for these zero-shot guards the unsafe-probability distribution is clustered, so the dev-calibrated threshold sits on a knife-edge and its G0/D1 catch count is **not reproducible** — it swung by >50 rows between library versions of the quantile routine. That fragility is itself a finding: **naive threshold transfer is unreliable for these guards**, so AP (ranking) and Δ_{context} (invariance) are the metrics we report. Behavior is heterogeneous: **Qwen3-4B** ranks best (AP·D 0.85) *and* is invariant on the protected minimal pairs ($\Delta_{\text{context}} \approx 0$), while **Qwen2.5-1.5B** shows a large protected-attribute score gap (Δ_{context} 0.18, max 0.51) — the minimal-pair gate discriminates guards that AP alone does not. AP·G rests on only 6 G1 positives in this split and is correspondingly noisy. These are **small-sample, LLM-judge-labeled, single-split diagnostics** (3 protected pairs): the benchmark is designed to *surface* these behaviors, not to certify any number as a fair-lending fact.

7. What this benchmark does *not* establish

- **Not SME-validated.** Labels are LLM-judge / policy-card-consistent. No fair-lending headline about a real lender or model is warranted until a stratified SME adjudication (especially of the G0/D1 rows and protected pairs) is recorded.
- **Not a full 2×2.** G1/D0 is empty; the orthogonality of G and D is demonstrated on three quadrants.
- **Not a regenerable dataset.** Generation is intentionally frozen; only the *evaluation* reproduces.
- **Not decontaminated against v2.** Cross-source near-dup removal vs the v2 general sources is pending.

8. Validation protocol (to reach confirmatory use)

1. **SME adjudication** of a stratified subset; upgrade cards from `sme_signoff: false`; report Fleiss- κ per label.
2. **Populate G1/D0** with curated domain-independent-harm-in-mortgage-clothing to complete the 2x2.
3. **Re-decontaminate** against the v2 general sources with MinHash near-dup removal.
4. **Add a domain-fine-tuned guard arm** (retrain the LoRA guards) to quantify the domain-adapter gain — and, jointly with the composition study, whether composing a general base with a mortgage adapter recovers general transfer.
5. **Select a license**; resolve the contested disparate-impact card (D07) with version-dating.

9. Conclusion

We release a fixed, HMDA-grounded, dual-labeled mortgage guardrail benchmark and a reproducible evaluation harness, and show baselines for how general and off-the-shelf guards behave on it. The benchmark’s value is the **G0/D1 stratum** — mortgage-policy violations that read as safe — and the **protected minimal pairs** that test guard invariance. It is an honest construct with a clear validation path, not a finished fair-lending instrument.

References

- Bowen, Price, Stein & Yang (2024). *Measuring and Mitigating Racial Bias in Large Language Model Mortgage Underwriting*. (grounding methodology; borderline / counterfactual design)
- Companion study: *The Benchmark Chooses the Winner* (Paper A) and *Compose, Don’t Tune?* (Paper B) in this repository.
- HMDA 2022 Snapshot National Loan-Level Dataset, FFIEC/CFPB.
- Llama Guard; WildGuard; ToxicChat; XSTest; OR-Bench (baseline guards / related benchmarks).